# Final Project Directions: Multi-Turn RL for Financial Dialogue Agents

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**Purpose of this meeting.** I would like feedback on one lead direction and a handful of alternates, all built around a quantitative and financial angle, before our team commits for the Nov 5 proposal. The specific questions I am hoping to settle are in Section 4.

## 1. Lead idea: learning when to ask, think, and act in a brokerage environment

**Anchored in:** L3 (dialogue as a POMDP), L12 (evaluation and user simulation), L15 (reasoning and test-time compute), L16 (agents, tool use, agentic RL), and Unit 7 (dialogic deference).

### Thesis

> *A small model trained with multi-turn reinforcement learning in a brokerage environment, where questions cost time and money and suitability rules make some of them mandatory, learns a better policy for when to ask, think, and act than prompting or distillation from frontier models. It resolves what it can itself, asks only what the client alone knows, stays compliant, and carries this over to clients it was not trained against.*

### Why a financial setting

Finance gives the problem two structural features that other domains would have to invent:

- **Regulation already classifies questions.** FINRA Rule 2111 requires reasonable diligence to ascertain a customer's investment profile (age, financial situation, tax status, objectives, time horizon, liquidity needs, risk tolerance, and so on); Reg BI carries a similar care obligation for retail recommendations. So each gap in the agent's information is either *mandatory* to ask, *redundant* (on file or derivable), *client-only* (a preference only the client holds), or *irrelevant*.
- **Asking has a dollar price.** Prices move between turns in the simulator, so every round trip to the client costs market exposure, while a wrong guess costs a policy violation or a bad fill. The end state (the final portfolio) is checked exactly, with no LLM judge in the reward.

### Gap against closest prior work

CollabLLM (Wu et al., ICML 2025) trains an 8B model to collaborate with simulated users using a multiturn-aware reward, but uses PPO/DPO with a GPT-4o-mini user simulator in the loop, no tools, no policy constraints, and mostly proxy end states; it also lists redundant questions as a weakness. “Ask or Assume?” (2026) and Ambig-SWE (2025) study clarification in coding agents, and UserBench (2025, as cited in L16) finds agents uncover fewer than 30% of unstated preferences; these largely address *under*-asking and do not separate information the agent could find itself from information only the user holds. τ-bench (Yao et al., 2024) has the right structure (hidden goal, policy document, end-state check) but is an evaluation benchmark rather than a training environment. To my knowledge no work trains the ask/think/act decision with RL in a policy-constrained, tool-using setting with exactly checkable outcomes, or ties clarification to a regulatory classification of questions. A fuller literature check will follow.

### Research questions

| Question | Hypothesis | Refuted if |
|---|---|---|
| **RQ1 (main).** Can multi-turn GRPO on a 1.5 to 3B model learn an ask / think / look-up / act policy? | Beats a prompted small model and SFT distillation from a frontier model on success, pass^k, and dollar cost | No gain over SFT distillation, or gains come only with more policy violations |
| **RQ2.** What did training change? | Gains decompose into fewer redundant questions, retained mandatory questions, and better per-turn allocation of thinking | Gains come from a single shortcut (e.g., never asking) rather than calibrated behavior |

| Question | Hypothesis | Refuted if |
|---|---|---|
| **RQ3.** Does it transfer? | A policy trained against a cheap scripted client holds up against LLM-simulated clients, including noisy ones | Gains vanish or reverse with LLM clients |

### Method

- **Environment.** Tools: `get_positions, get_quote, risk_check, place_order, ask_client`. A policy document sets position limits, a restricted list, order-size and margin rules. Tasks (e.g., "rebalance to 60/40 and trim tech") are generated with a known mix of information classes, so every question the agent asks can be labeled needed or unneeded.
- **Clients.** Training uses a fast, deterministic scripted client that answers truthfully from the hidden goal; this is what makes RL affordable. Evaluation adds LLM-simulated clients and noisy variants (terse, verbose, occasionally mistaken). No human subjects.
- **Reward.** End-state success against the hidden goal and policy, minus a per-question cost tied to simulated market movement, minus a large penalty per violation.
- **Training.** GRPO with LoRA on a Qwen-class 1.5 to 3B model, using an existing multi-turn RL library (OpenPipe ART, *verifiers*, or verl, to be confirmed on PACE-ICE). Curriculum from fully specified to underspecified tasks.
- **Baselines.** Same model prompted; same model SFT-distilled from filtered frontier-model trajectories; a prompted frontier model as a ceiling.
- **Ablations.** Question price (zero / low / high); thinking off vs on vs per-turn; look-up tools removed.
- **Evaluation.** Multi-turn success, pass^k, dollars lost, turns, needed vs unneeded question rates, missed mandatory questions; 3 seeds with confidence intervals; error analysis on real transcripts.

### Compute and timeline

Rough estimate: a 3B model with LoRA, about 6 turns and 2k tokens per episode, 32 tasks × 8 rollouts per GRPO step, giving roughly 5 to 10 GPU-hours per run on one A100/H100. Three seeds across the main run and two to three ablations is about **50 to 100 GPU-hours** on PACE-ICE. Baselines and LLM-client evaluation need modest API spend.

| Window | Work |
|---|---|
| Oct 28 to Nov 5 | Environment spec, tooling smoke test on ICE, proposal |
| Nov 5 to Nov 19 | Environment, scripted client, policy checker; prompted baselines; first GRPO run on a small model (milestone) |
| Nov 19 to Dec 1 | Main runs and seeds, SFT baseline, ablations, LLM-client transfer evaluation |
| Dec 1 to Dec 8 | Poster (Dec 3), error analysis, final paper |

### Risks and fallback

- **No learning signal.** Checked by week 2 on easy tasks with a small model; curriculum and reward shaping as remedies.
- **Reward hacking** (never acting, or asking endlessly). Detected via question rates on fully specified tasks and violation counts.
- **Fallback that still yields a paper.** The environment plus a measurement study of how frontier models price questions in it, and how they allocate thinking across turns, needs only API calls and is a complete contribution if RL underperforms.

## 2. Other directions with a quantitative angle (in case the lead is out of scope)

| Idea | Research question | Lectures | Main risk |
|---|---|---|---|
| **A. Earnings-call Q&A** | Does turn-level evasion under analyst follow-up pressure carry signal, and does audio or a latent probe add to text? | L18–21, L24 | Audio data access; memorized market reactions |
| **B. Fed press conferences** | Same question on public-domain FOMC audio and transcripts, with a cleaner event-time market response | L18–21, L24 | Small sample (about eight per year) |
| **C. RFQ negotiation** | Does an explicit belief state over a counterparty's reservation price help an LLM dealer negotiate and leak less? | L3, Unit 7 | Simulator realism |
| **D. Per-turn thinking** | In a multi-turn financial task, which turns need reasoning, and can a cheap router match always-think at a fraction of tokens? | L15, L16 | Overlap with single-turn adaptive thinking |
| **E. Multi-turn faithfulness** | When an offhand client remark several turns earlier shifts a computed recommendation, does the reasoning admit it? | L15 | May already exist; needs literature check |
| **F. Tool-grounded sycophancy** | Do models abandon correct, computable quant answers (bond math, option pricing) under pushback, and does tool use protect them? | Unit 7, L16 | Could read as a small result |
| **G. Numeric drift** | Over a long stateful portfolio session, do in-chain arithmetic errors compound, and do tool calls reset or relocate them? | L14, L16 | Narrow scope |

A cross-cutting caution for anything predictive: models have memorized historical prices and news, so results on past market data risk measuring recall; simulation or post-cutoff data avoids this. D and F also work as fallbacks inside the lead environment.

## 3. What would make this a strong workshop paper

Three contributions that stand independently (the environment with labeled information classes, the measurement of current models in it, and the RL result with its behavioral breakdown); a non-straw baseline (SFT distillation from a frontier model) with a clear refutation condition for each RQ; and results disaggregated by client type and information class.

## 4. Questions for you

1. Does the lead idea pass the “open” test as framed, and is multi-turn RL realistic in five weeks?
2. Is 50 to 100 GPU-hours on PACE-ICE achievable for a course team, or is there a better source of compute?
3. Is a scripted client for training plus LLM-simulated clients for evaluation an acceptable user-simulation design, given the human-subjects constraint?
4. Is the financial and regulatory framing a strength for a SIGDIAL or NLP4ConvAI-style venue, or would you push for a more domain-general setting?
5. If not the lead idea, which alternate would you steer us toward?

## Selected references


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